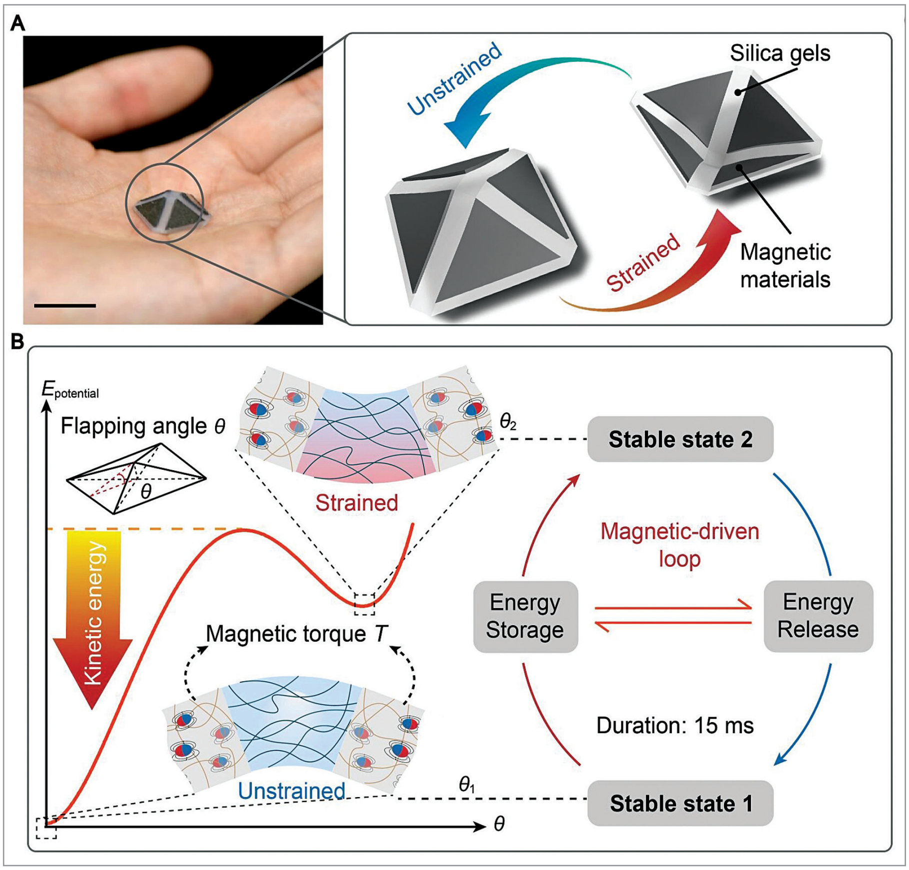
# RESEARCH
## & INNOVATION UPDATES

# Magnetically driven soft jumping robot

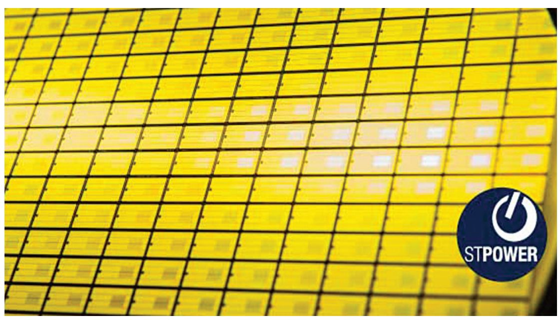



Researchers from Zhejiang University in China have developed a magnetically driven, bistable soft jumper with superior jumping capabilities. This robot jumps higher and faster than previous models, offering potential applications in navigating complex terrains and confined environments. Using elastic, deformable materials, it resists impact damage while achieving take-off speeds over two metres per second and response times under 15 milliseconds. The bistable mechanism releases stored energy, enabling jumps more than 108 times its body height. Tests showed it can navigate amphibious terrain, U-shaped structures, and jump from underwater. Adjustable magnetic fields control jump height and distance, making it versatile for future robotic systems.

*Schematic illustration of bistable soft jumper (Credit: Daofan Tang)*

## SiC MOSFET technology for EV traction inverters

STMicroelectronics has introduced its fourth-generation STPOWER silicon carbide (SiC) MOSFET technology, setting new standards in power efficiency, density, and durability for automotive and industrial applications. Optimised for electric vehicle (EV) traction inverters, these SiC MOSFETs are available in 750V and 1200V versions, improving EV performance with faster charging, lighter designs, and extended range. The technology also enhances high-power industrial applications like solar inverters and data centres, delivering higher energy efficiency and lower operational costs. The company plans to release a fifth generation by 2027, further advancing e-mobility and industrial solutions.




*4th generation SiC MOSFETs (Credit: https://newsroom.st.com)*